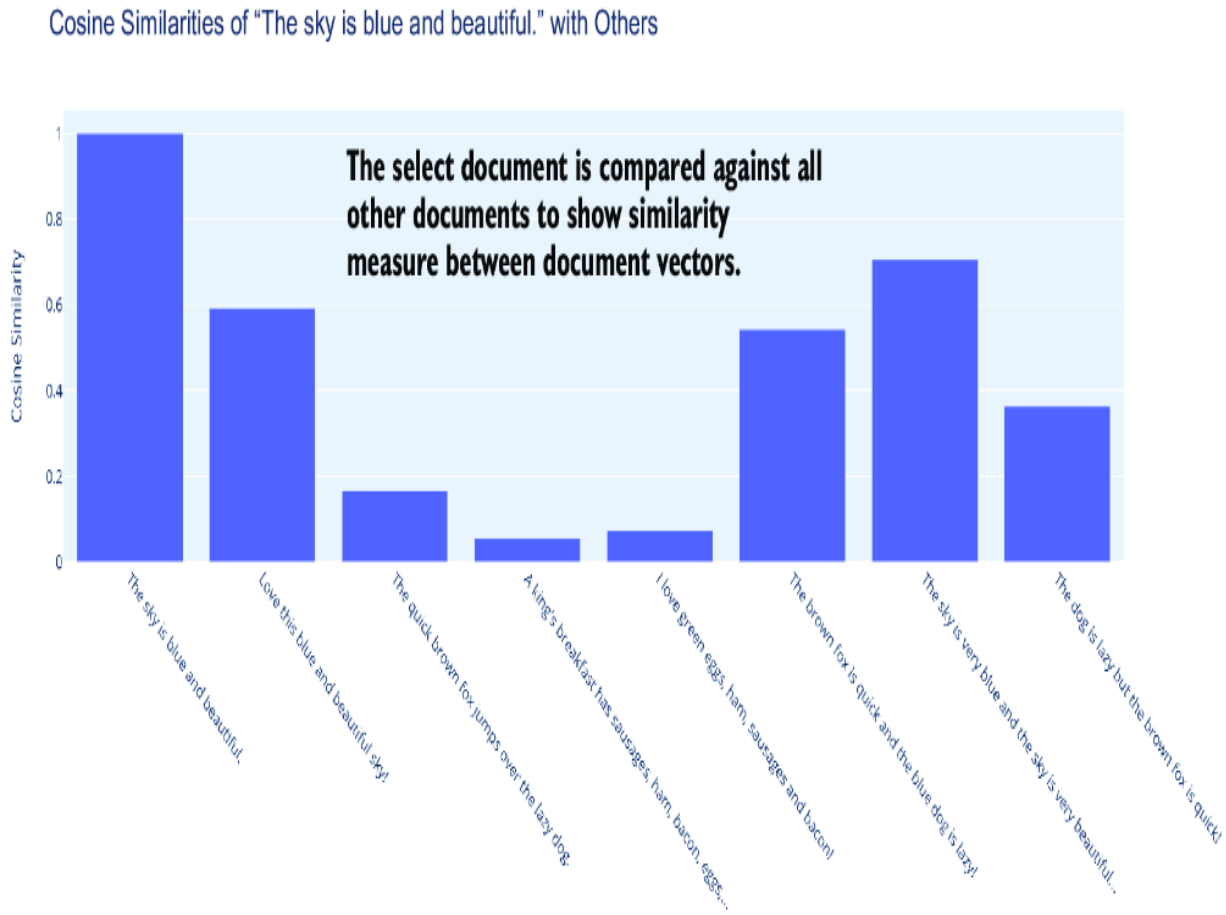


Figure 8.5 The cosine similarity between selected documents and the document set

The method of vectorization will dictate the measure of semantic similarity between documents. Before we move on to better methods of vectorizing documents, we'll examine storing vectors to perform vector similarity searches.

8.3.2 Vector databases and similarity search

After vectorizing documents, they can be stored in a vector database for later similarity searches. To demonstrate how this works, we can efficiently replicate a simple vector database in Python code.

Open `document_vector_database.py` in VS Code, as shown in listing 8.3. This code demonstrates creating a vector database in memory and then allowing users to enter text to search the database and return results. The results returned show the document text and the similarity score.